OOAD Lab 1: Case Study on Use Case Diagram

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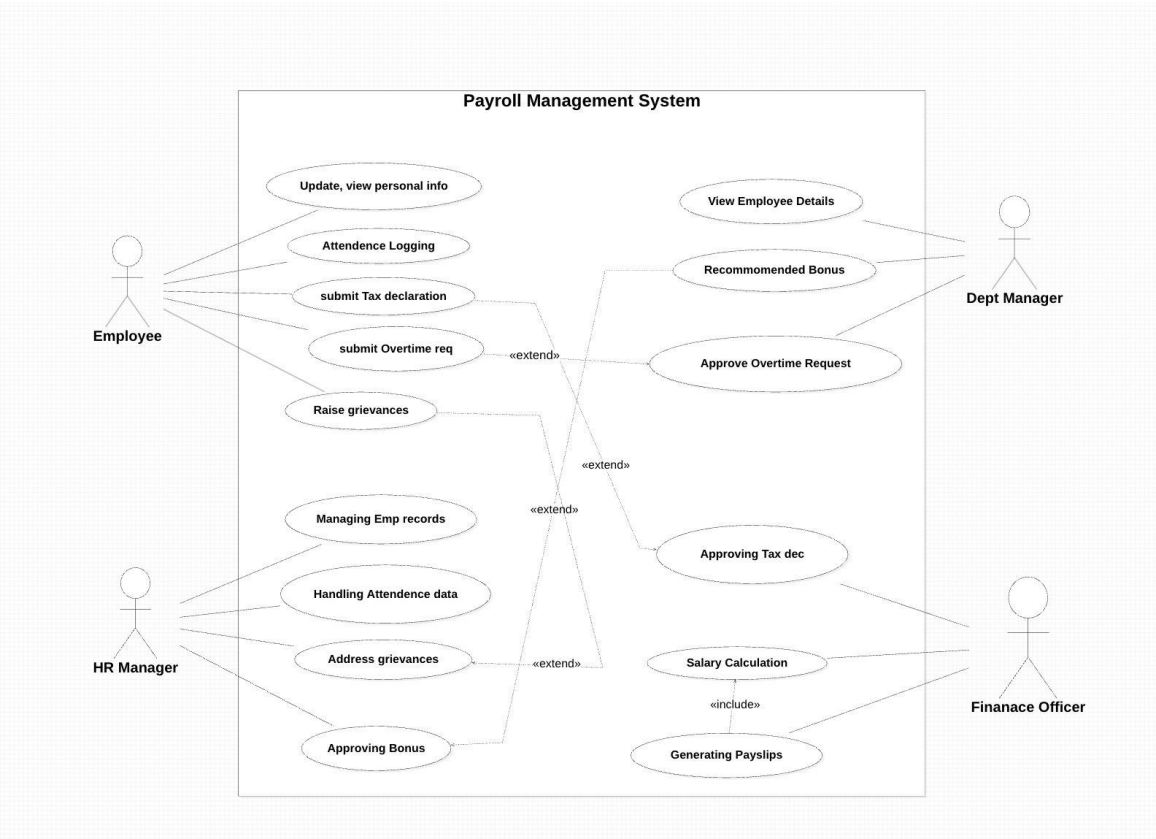
AIML 6th SEM D SECTION

Case Study 1: Payroll Management System



Use Case 1: Salary Calculation

Field	Description
Use Case Name	Salary Calculation
Primary Actor	Finance Officer
Description	Calculates the final salary of an employee using attendance data, approved overtime requests, approved bonuses, and approved tax declarations.
Preconditions	Finance Officer is logged in. Attendance is recorded. Overtime requests are approved. Bonus recommendations are approved. Tax declarations are approved.
Postconditions	Employee's gross and net salary are calculated and stored in the system.

Main Flow

1. Finance Officer logs in.
2. Finance Officer selects an employee for salary processing.
3. System retrieves attendance data.
4. System retrieves approved overtime request details.
5. System retrieves approved bonus details.
6. System retrieves approved tax declarations.
7. System calculates the gross salary.
8. System applies tax and other deductions.
9. System stores the final calculated salary.

Alternate Flow

- If any required data is missing or not approved, the system displays an error message and stops the salary calculation process.

Use Case 2: Generating Payslips

Field	Description
Use Case Name	Generating Payslips
Primary Actor	Finance Officer
Description	Generates an official payslip for an employee after successful salary calculation.
Preconditions	Finance Officer is logged in. Salary has already been calculated.
Postconditions	Payslip is generated, stored in the system, and becomes available to the employee.

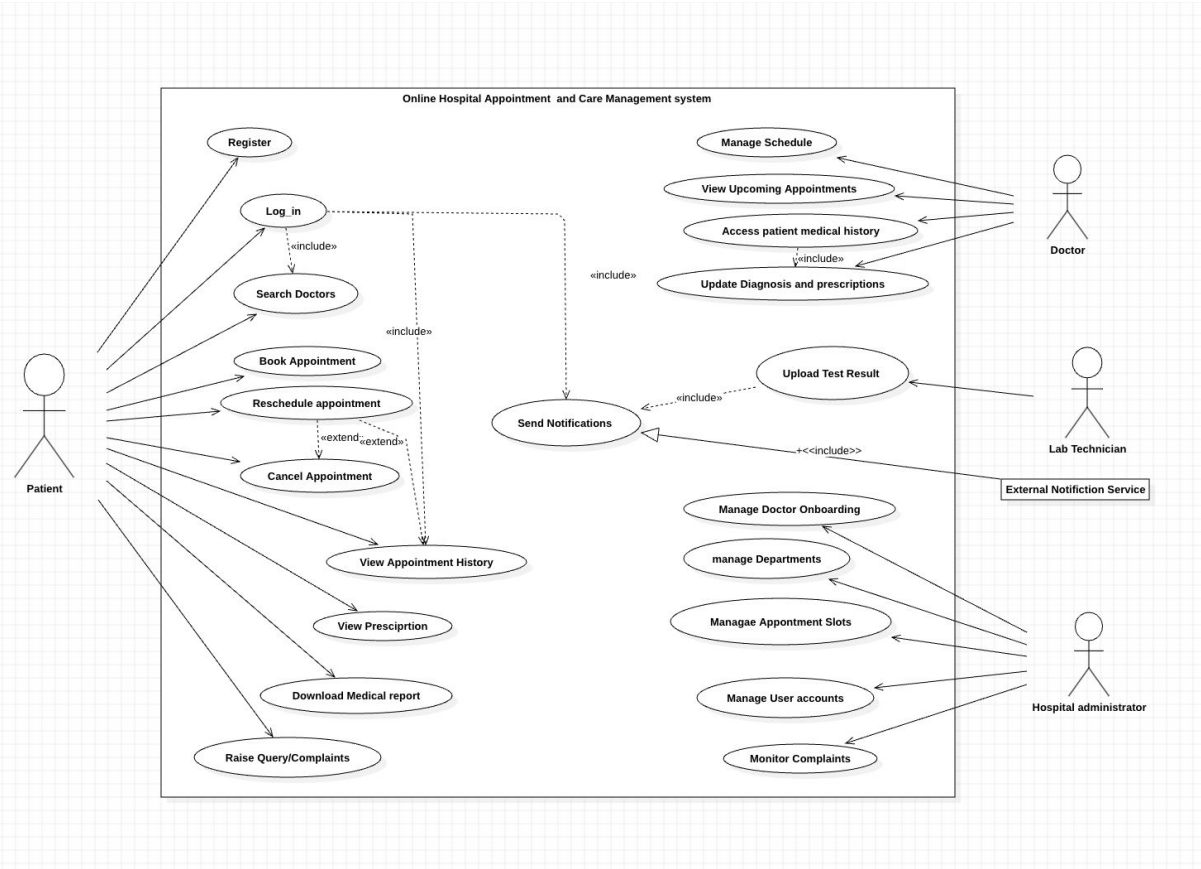
Main Flow

1. Finance Officer selects an employee.
2. System verifies whether salary calculation is completed.
3. System generates the payslip.
4. Payslip is stored in the system.
5. Employee can view or download the payslip.

Alternate Flow

- If salary is not calculated, the system blocks payslip generation and displays an appropriate message.

Case Study 2: Online Hospital Appointment & Care Management



Use Case 1: Book Appointment

Field	Description
Use Case Name	Book Appointment
Primary Actor	Patient
Description	Allows a patient to search for doctors and book, reschedule, or cancel appointments based on availability.
Preconditions	Patient is registered and logged in. Doctors and their appointment slots exist in the system.
Postconditions	Appointment is created, updated, or cancelled. Doctor's schedule is updated. Patient receives a notification.

Main Flow

1. Patient logs in to the system.
2. Patient searches for doctors by specialization, department, or availability.
3. System displays matching doctors along with available time slots.
4. Patient selects a doctor and preferred time slot.
5. System verifies the availability of the selected slot.
6. Appointment is booked.
7. Doctor's schedule is updated.

8. Confirmation notification is sent to the patient.

Alternate Flow

- If the selected slot is unavailable, the patient chooses another slot.
- If the patient reschedules or cancels the appointment, the system updates the appointment and notifies both the patient and doctor.

Use Case 2: Manage Consultation and Medical Records

Field	Description
Use Case Name	Manage Consultation and Medical Records
Primary Actors	Doctor, Lab Technician
Description	Handles patient consultation, diagnosis updates, prescription management, and uploading of medical test reports.
Preconditions	A valid appointment exists. Doctor and patient are logged in.
Postconditions	Diagnosis and prescriptions are saved. Lab reports are uploaded and linked to the patient profile. Patient can access updated medical records.

Main Flow

1. Doctor views scheduled appointments.
2. Doctor selects a patient.
3. System displays the patient's medical history.
4. Doctor enters diagnosis and prescription details.
5. Consultation details are saved in the system.
6. Lab technician uploads test results.
7. Uploaded reports are linked to the patient profile.
8. Patient is notified about report availability.

Alternate Flow

- If no lab tests are required, the process ends after prescription entry.
- If reports are delayed, notification is sent only after report upload.